

Tutorial 1

ECON 101 – Macroeconomics

Fall 2025

Part A. Numerical Problems

Q1. Market Equilibrium and Price Controls

The market for coffee is:

$$Q_d = 100 - 2P, \quad Q_s = 20 + 3P.$$

- (a) Find the equilibrium price and quantity.
- (b) With a price floor of $P = 20$, compute the surplus (if any).
- (c) With a price ceiling of $P = 10$, compute the shortage (if any).

Q2. Demand Shift from Income

Demand for laptops is

$$Q_d = 500 - 5P + 0.5Y$$

and supply is

$$Q_s = 100 + 3P.$$

- (a) If $Y = 200$, find the equilibrium P and Q .
- (b) If $Y = 300$, find the new equilibrium P and Q .
- (c) Briefly interpret how the income change affects equilibrium.

Q3. Point Elasticity and Revenue

Demand is

$$Q_d = 200 - 4P.$$

- (a) Compute the price elasticity of demand at $P = 30$.
- (b) Classify demand at that price (elastic, inelastic, unit elastic).
- (c) What happens to total revenue if price increases slightly from $P = 30$?

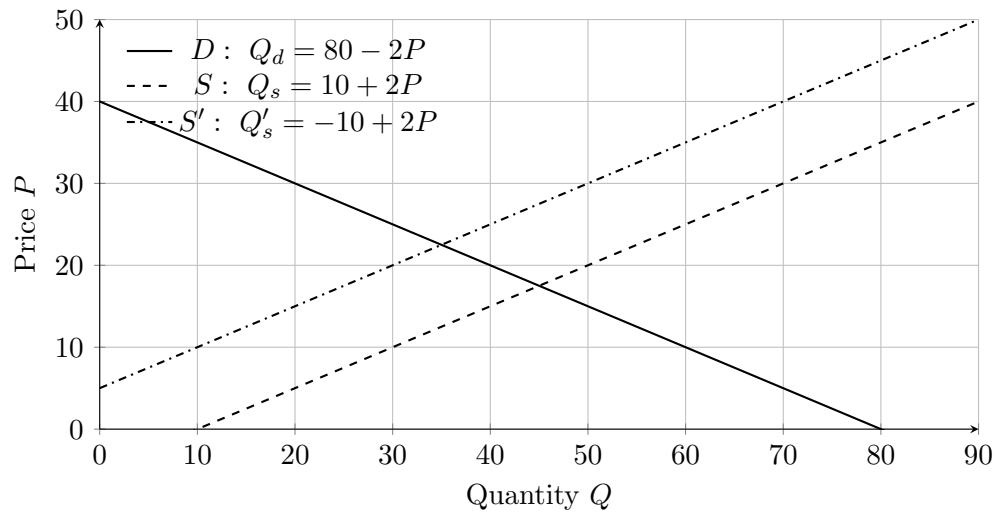
Part B. Graphical Problems

Q4. Supply Shock (Graphical and Numerical)

Initial demand and supply:

$$Q_d = 80 - 2P, \quad Q_s = 10 + 2P.$$

- (a) Graph Q_d and Q_s ; mark the initial equilibrium.
- (b) An input cost increase shifts supply to $Q'_s = -10 + 2P$. Graph Q'_s on the same axes and identify the new equilibrium.
- (c) State the direction of the changes in equilibrium P and Q and why.

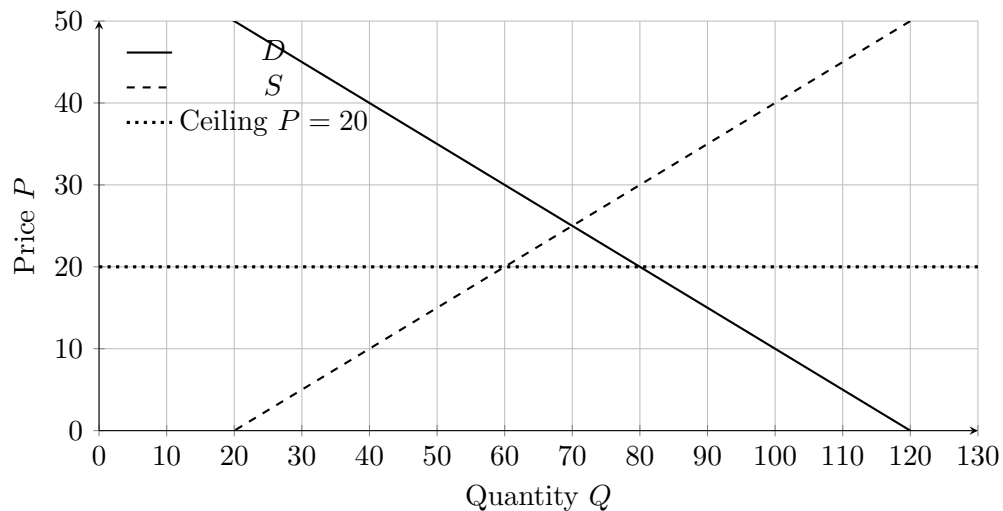


Q5. Rent Ceiling (Graphical and Numerical)

Rental housing market:

$$Q_d = 120 - 2P, \quad Q_s = 20 + 2P.$$

- (a) Graph D and S and mark the competitive equilibrium.
- (b) Impose a rent ceiling at $P = 20$. On the graph, show the shortage.
- (c) Compute the numerical size of the shortage.

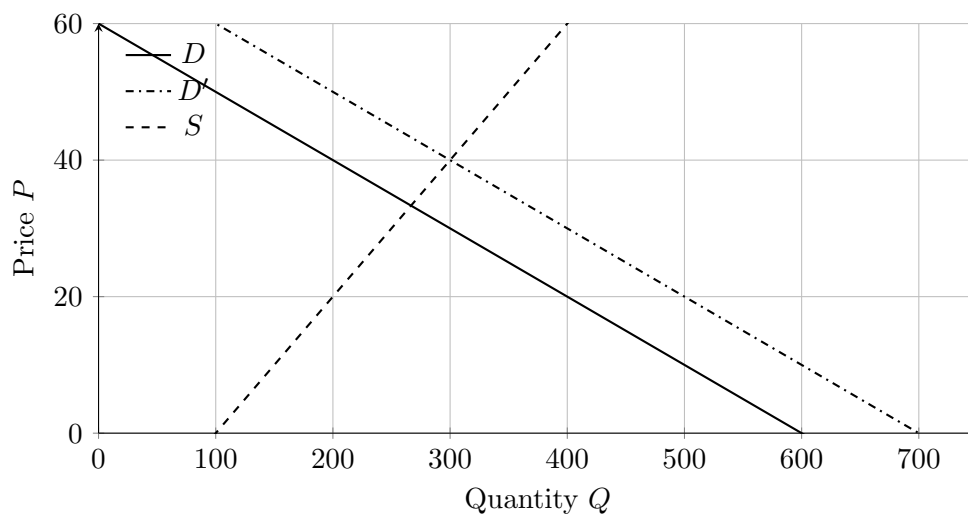


Q6. Comparative Statics: Demand Increase

Wheat market:

$$Q_d = 600 - 10P, \quad Q_s = 100 + 5P.$$

- Solve for the initial equilibrium.
- Demand increases to $Q'_d = 700 - 10P$. Find the new equilibrium.
- On a diagram, show the shift and both equilibria. Briefly explain the changes.



Solutions

Q1.

- (a) $100 - 2P = 20 + 3P \Rightarrow 80 = 5P \Rightarrow P^* = 16, Q^* = 68.$
- (b) At $P = 20$: $Q_d = 100 - 40 = 60, Q_s = 20 + 60 = 80 \Rightarrow \text{surplus} = 20.$
- (c) At $P = 10$: $Q_d = 100 - 20 = 80, Q_s = 20 + 30 = 50 \Rightarrow \text{shortage} = 30.$

Q2.

- (a) $Y = 200 \Rightarrow Q_d = 500 - 5P + 100 = 600 - 5P.$
 $600 - 5P = 100 + 3P \Rightarrow 500 = 8P \Rightarrow P^* = 62.5.$
 $Q^* = 100 + 3(62.5) = 287.5.$
- (b) $Y = 300 \Rightarrow Q_d = 500 - 5P + 150 = 650 - 5P.$
 $650 - 5P = 100 + 3P \Rightarrow 550 = 8P \Rightarrow P^{*'} = 68.75.$
 $Q^{*'} = 100 + 3(68.75) = 306.25.$
- (c) Income rises, demand shifts right. For a normal good, both P and Q increase.

Q3.

- (a) $Q = 200 - 4P.$ At $P = 30, Q = 80.$
 $\frac{dQ}{dP} = -4.$ Elasticity $\varepsilon = \left(\frac{dQ}{dP}\right) \left(\frac{P}{Q}\right) = (-4) \left(\frac{30}{80}\right) = -1.5.$
- (b) $|\varepsilon| = 1.5 > 1 \Rightarrow$ demand is elastic at $P = 30.$
- (c) When demand is elastic, a price increase reduces total revenue; TR falls from a marginal increase in P at $P = 30.$

Q4.

- (a) $80 - 2P = 10 + 2P \Rightarrow 70 = 4P \Rightarrow P^* = 17.5, Q^* = 80 - 2(17.5) = 45.$
- (b) With $Q'_s = -10 + 2P$: $80 - 2P = -10 + 2P \Rightarrow 90 = 4P \Rightarrow P^{*'} = 22.5, Q^{*'} = 80 - 2(22.5) = 35.$
- (c) Higher input costs shift supply left, raising P and reducing $Q.$

Q5.

- (a) $120 - 2P = 20 + 2P \Rightarrow 100 = 4P \Rightarrow P^* = 25, Q^* = 70.$
- (b) With a ceiling at $P = 20$: $Q_d = 120 - 40 = 80, Q_s = 20 + 40 = 60.$
- (c) Shortage = $Q_d - Q_s = 20.$

Q6.

- (a) $600 - 10P = 100 + 5P \Rightarrow 500 = 15P \Rightarrow P^* = \frac{500}{15} \approx 33.33, Q^* = 600 - 10(33.33) \approx 266.67.$
- (b) With $Q'_d = 700 - 10P$: $700 - 10P = 100 + 5P \Rightarrow 600 = 15P \Rightarrow P^{*'} = 40, Q^{*'} = 700 - 10(40) = 300.$
- (c) Rightward demand shift raises both equilibrium price and quantity as buyers are willing to purchase more at each price.